[39] Vittorio Girotto, Andrea Mazzocco, and Alessandro Tasso. The effect of premise order in conditional reasoning: A test of the mental model theory. *Cognition*, 63(1):1–28, 1997. doi: 10.1016/S0010-0277(96)00792-5. URL https://pubmed.ncbi.nlm.nih.gov/9187063/.

[40] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. Knowledge distillation: A survey. *International Journal of Computer Vision*, 129(6):1789–1819, 2021.

[41] Manish Gupta and Puneet Agrawal. Compression of deep learning models for text: A survey. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 16(4):1–55, 2022.

[42] Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao, and Tianyi Zhou. A survey on knowledge distillation of large language models, 2024. URL https://arxiv.org/abs/2402.13116.

[43] DeepSeek-AI. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning, 2025. URL https://arxiv.org/abs/2501.12948.

[44] Ahmed El-Kishky, Daniel Selsam, Francis Song, Giambattista Parascandolo, Hongyu Ren, Hunter Lightman, Hyung Won Chung, Ilge Akkaya, Ilya Sutskever, Jason Wei, Jonathan Gordon, Karl Cobbe, Kevin Yu, Lukas Kondraciuk, Max Schwarzer, Mostafa Rohaninejad, Noam Brown, Shengjia Zhao, Trapit Bansal, Vineet Kosaraju, Wenda Zhou, and OpenAI. Learning to Reason with LLMs, 2024. URL https://openai.com/index/learning-to-reason-with-llms/.

[45] Fengli Xu, Qianyue Hao, Zefang Zong, Jingwei Wang, Yunke Zhang, Jingyi Wang, Xiaochong Lan, Jiahui Gong, Tianjian Ouyang, Fanjin Meng, Chenyang Shao, Yuwei Yan, Qinglong Yang, Yiwen Song, Sijian Ren, Xinyuan Hu, Yu Li, Jie Feng, Chen Gao, and Yong Li. Towards large reasoning models: A survey of reinforced reasoning with large language models, 2025. URL https://arxiv.org/abs/2501.09686.

[46] Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational scaling laws and the predictability of langauge model performance. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL https://openreview.net/forum?id=On5WIN7xyD.

A A formal presentation of the Erotetic Theory

ETR is formulated in [10] in a set-theoretic language, and we reproduce the full definitions here, with the explicit permission of the authors. We remark that Definitions 1 and 11 contain the parameters of the theory, i.e. the basic atomic propositions, the constants and function symbols, and a set logical and extralogical axioms. The central definition is that of a *View*, Definition 9. Following that are definitions of the basic operations of ETR, in particular Update (Definition 23) and Query (Definition 30) whose use in PyETR were demonstrated in Section 2.

Definition 1 (**Basic objects**).
Let $\mathbf{A}, \mathbf{F}, \{?\}$, $\mathbf{P}^{\top}$, $\mathbf{P}^{\bot}$ *be pairwise disjoint countable sets.*

($\mathbf{F}$ *arity)* $\alpha : \mathbf{F} \to \mathbb{N}$. *Write* $f \in \mathbf{F}$ *as* f^k *when* $\alpha(f) = k$.
(Constants) $\exists w_{\in \mathbf{F}}\ \alpha(w) = 0$.
($\mathbf{P}$ *polarity)* *Let* $N : \mathbf{P}^{\top} \to \mathbf{P}^{\bot}$ *be a bijection. Write* $\bar{P}$ *for* $N(P_{\in \mathbf{P}^{\top}})$.
Let $\mathbf{P} = \mathbf{P}^{\top} \cup \mathbf{P}^{\bot}$.
($\mathbf{P}$ *arity)* $\alpha' : \mathbf{P} \to \mathbb{N}$. *Write* P *as* P^k *for* $\alpha'(P) = k$. $\alpha'(P) = \alpha'(\bar{P})$.
(Identity) $=^2 \in \mathbf{P}$.

Definition 2 (**Terms T**).
$\mathbf{A} \subseteq \mathbf{T}$. *If* $f^0 \in \mathbf{F}$, *then* $\langle f^0 \rangle \in \mathbf{T}$. *If* $f^k \in \mathbf{F} \wedge \{t_1 \dots t_k\} \subseteq \mathbf{T}$, *then* $\langle f^k, \langle t_1 \dots t_k \rangle \rangle \in \mathbf{T}$.

Definition 3 (**Atoms** $\mathcal{A}$).
If $P^k \in \mathbf{P}$ *and* $\vec{\tau} = \langle t_1 \dots t_k \rangle$ *for* $t_i \in \mathbf{T}$, *then* $\langle P^k, \vec{\tau} \rangle \in \mathcal{A}$ *(abbrv.* $P\vec{\tau}$*).* $|\vec{\tau}| = \{t_1 \dots t_k\}$.

Definition 4 (**States** $\mathbb{S}$).
$\mathbb{S} = \mathcal{P}(\mathcal{A})$. *Abbreviate '{ }' as '0'.*

Definition 5 (**Dependency relations** $\mathcal{R}$).
Let $\Gamma \in \mathcal{P}(\mathbb{S})$. *Let* $U, E \subseteq \mathbf{A}(\Gamma)$ *and* $D \subseteq E \times U = \{\langle e, u \rangle : e \in E \wedge u \in U\}$. *Define* $\langle U, E, D \rangle \in \mathcal{R}_{\Gamma}$ *iff*

(Bipartite) $U \cap E = \emptyset \wedge D \subseteq E \times U$, *and*
(Matryoshka) *For all* $u, u' \in U$ $\{e \in E : \langle e, u \rangle \in D\} \subseteq \{e \in E : \langle e, u' \rangle \in D\}$ *or*
$\{e \in E : \langle e, u' \rangle \in D\} \subseteq \{e \in E : \langle e, u \rangle \in D\}$.

For $R \in \mathcal{R}$ *write* $R = \langle U_R, E_R, D_R \rangle$. *For* $\langle \emptyset, \emptyset, \emptyset \rangle$ *write* $0_{\mathcal{R}}$.

Definition 6 (**Open terms** $\mathbf{T_1}$**and atoms** $\mathcal{A}_1$).
$? \in \mathbf{T}_1$ *and* $\langle f^k, \langle t_1, \dots, t_k \rangle \rangle \in \mathbf{T}_1$ *whenever* $f^k \in \mathbf{F}$ *and for some* $1 \leq i \leq k$, $t_i \in \mathbf{T}_1$ *and* $t_j \in \mathbf{T}$ *for* $j \neq i$. $\langle P^k, \langle t_1, \dots, t_k \rangle \rangle \in \mathcal{A}_1$, *whenever* $P^k \in \mathbf{P}$ *and for some* $1 \leq i \leq k$, $t_j \in \mathbf{T}$ *for all* $j \neq i$ *and* $t_i \in \mathbf{T}_1$.

Definition 7 (**Issue structures** $\mathbb{I}$).
$I \in \mathbb{I}_{\Gamma}$ *iff* I *consists of pairs* $\langle t, x \rangle$, *s.t.* $x \in \mathcal{A}_1$ *and* $x[t/?] \in \mathcal{A}(\Gamma)$.

Definition 8 (**Issue matches** M_{IJ}).
For $I, J \in \mathbb{I}_{\Gamma}$, *define,*

$$M_{IJ} = \{\langle t_1, t_2 \rangle : \exists x(\langle t_1, x \rangle \in I \wedge (\langle t_2, x \rangle \in J \vee \langle t_2, \bar{x} \rangle \in J))\}$$

where, for $x = \langle P^k, \langle t_1, \dots, t_k \rangle \rangle$, $\bar{x} = \langle \bar{P}^k, \langle t_1, \dots, t_k \rangle \rangle$.

Definition 9 (**Views** $\mathbb{V}$).
For Γ, Θ *finite subsets of* $\mathbb{S}$, $R \in \mathcal{R}_{\Gamma \cup \Theta}$, $I \in \mathbb{I}_{\Gamma \cup \Theta}$,
$\langle \Gamma, \Theta, R, I \rangle \in \mathbb{V}$ *(abbreviated* $\Gamma^{\Theta}_{RI} \in \mathbb{V}$*).*
Write $\top$ *for* $\{0\}^{\{0\}}_{0_{\mathcal{R}}\emptyset}$ *and* $\bot$ *for* $\emptyset^{\{0\}}_{0_{\mathcal{R}}\emptyset}$.

Definition 10 (**Commitments** $\mathbb{C}$).
For $C \subseteq \mathbb{V}, G \in \mathbb{V}$, $\langle C, G \rangle \in \mathbb{C}$.

Definition 11 (**Primitive absurd states** $\mathbb{K}$).
Let $\mathbb{K}_{\subseteq \mathbb{S}}$ *contain,* $\forall t, t'_{\in \mathbf{T}} \forall p_{\in \mathcal{A}} \forall x_{\in \mathcal{A}_1}$, *at least:*

(LNC) $\{p, \bar{p}\}$,
(Aristotle) $\{\neq tt\}$,

(Leibniz) $\{=tt', x[t/?], \bar{x}[t'/?]\}$.

Definition 12 ($\mathcal{R}$ **Restriction).**
$[R]_X = \langle U_R \cap X, E_R \cap X, D_R \cap ((E_R \cap X) \times (U_R \cap X))\rangle$.
$[R]_\Gamma = [R]_{\mathbf{A}(\Gamma)}$.
Given $\langle \Gamma, \Theta, R, I\rangle$*, we allow ourselves to write* $\langle \Gamma, \Theta, [R], I\rangle$ *for* $\langle \Gamma, \Theta, [R]_{\Gamma\cup\Theta}, I\rangle$.

Definition 13 ($\mathbb{I}$ **Restriction).**
Within a quadruple $\langle \Gamma, \Theta, R, [I]\rangle$*, let*
$[I] = \{\langle t, x\rangle : \langle t, x\rangle \in I \wedge x[t/?] \in \mathcal{A}(\Gamma \cup \Theta)\}$

Definition 14 ($\mathcal{R}$ **Algebra).**
Let $R \rtimes S = \langle U_R \cup U_S, E_R \cup E_S, D_R \cup D_S \cup E_S \times U_R\rangle$ *and let*
$0_\mathcal{R} \bowtie 0_\mathcal{R} = 0_\mathcal{R}$.
Let

$$R \bowtie S = \langle U_0, E_0, \emptyset\rangle \rtimes ([R]_{\mathbf{A}(R)-(E_0\cup U_0)} \bowtie [S]_{\mathbf{A}(S)-(E_0\cup U_0)})$$

where $E_0 = \{e_{\in E_R\cup E_S} : \forall u.\langle e, u\rangle \notin D_R \cup D_S\}$ *and*
$U_0 = \{u_{\in U_R\cup U_S} : \forall e \notin E_0.\langle e, u\rangle \notin (E_R \times U_R - D_R) \cup (E_S \times U_S - D_S)\}$.

Definition 15 (Product).
$\Gamma^\Theta \otimes \Delta^\Psi = \big(\{\gamma_{\in\Gamma} \cup \delta_{\in\Delta} : \exists\psi_{\in\Psi}(\psi \subseteq \gamma)\} \cup \{\gamma_{\in\Gamma} : \neg\exists\psi_{\in\Psi}(\psi \subseteq \gamma)\}\big)^\Theta$
$(\Gamma \otimes \Delta)^{\{0\}} = \Gamma^{\{0\}} \otimes \Delta^{\{0\}}$
$\bigotimes_{i\in\mathbf{P}} \Delta_i^{\Psi_i} = \{0\}^{\{0\}} \otimes \Delta_1^{\Psi_1} \otimes \ldots \otimes \Delta_n^{\Psi_n}$
$\Gamma^\Theta_{RI} \otimes^T \Delta^\Psi_{SJ} = (\Gamma^\Theta \otimes \Delta^\Psi)_{[(T\bowtie R)\bowtie(T\bowtie S)][I\cup J]}$
$\Gamma^\Theta_{RI} \otimes \Delta^\Psi_{SJ} = \Gamma^\Theta_{RI} \otimes^{0_\mathcal{R}} \Delta^\Psi_{SJ}$
$\bigotimes^T_{i\in\mathbf{P}} \Delta_i{}^{\Psi_i}_{S_iJ_i} = \top \otimes^T \Delta_1{}^{\Psi_1}_{S_1J_1} \otimes^T \ldots \otimes^T \Delta_n{}^{\Psi_n}_{S_nJ_n}$

Definition 16 (Sum).
$\Gamma^\Theta_{RI} \oplus^T \Delta^\Theta_{SJ} = (\Gamma \cup \Delta)^\Theta_{[(T\bowtie R)\bowtie(T\bowtie S)][I\cup J]}$
$\Gamma^\Theta_{RI} \oplus \Delta^\Theta_{SJ} = \Gamma^\Theta_{RI} \oplus^{0_\mathcal{R}} \Delta^\Theta_{SJ}$
$\bigoplus^T_{i\in\mathbf{P}} \Delta^{\Psi_i}_{S_iJ_i} = \emptyset^\Psi_{0_\mathcal{R}\emptyset} \oplus^T \Delta_1{}^\Psi_{S_1J_1} \oplus^T \ldots \oplus^T \Delta_n{}^\Psi_{S_nJ_n}$

Definition 17 ($\mathcal{A}$ **Answer potential).**
$\Gamma[\Delta]^{\mathcal{A}P} = |\mathcal{A}(\Gamma) \cap \mathcal{A}(\Delta)|$

Definition 18 (Answer).

$$\Gamma^\Theta_{RI}[\Delta^{\{0\}}_{SJ}]^A = (\arg\max_{\gamma\in\Gamma} \Delta[\{\{p\} : p \in \gamma\}]^{\mathcal{A}P})^\Theta_{[R][I]}$$

Definition 19 (Negation).
$[\Gamma^\Theta_{RI}]^N = (\Theta^{\{0\}} \otimes ([\Gamma]^N)^{\{0\}})_{[R]^N[I]^N}$
$[\Gamma]^N = \bigotimes_{\gamma\in\Gamma} \hat{\gamma}^{\{0\}}$
$\hat{\gamma} = \big\{\{\bar{p}\} : p_{\in\gamma}\big\}$
$\bar{p} = \bar{F}\vec{\tau}$ *if* $p = F\vec{\tau}$; $\bar{p} = F\vec{\tau}$ *if* $p = \bar{F}\vec{\tau}$
$[R]^N = \langle E_R, U_R, \{\langle a, b\rangle \in U_R \times E_R : \langle b, a\rangle \notin D_R\}\rangle$
$[I]^N = I \cup \{\langle t, \bar{x}\rangle : \langle t, x\rangle \in I\}$

Definition 20 (Novelty)**.** $a \in \mathbf{A}$ *is* novel *during an inference step* $\langle C, G\rangle[D]^O$ *if* $a \notin \mathbf{A}(G) \cup \mathbf{A}(D)$ *and* a *has not yet appeared in the computation of the conclusion of the step. A function* $\nu : X \to \mathbf{A}$*, where* $X \subseteq \mathbf{A}$ *is finite, is novel during an inference step if each* $\nu(x)$ *is novel and* ν *is injective, and then* $[\nu]_X$ *denotes the simultaneous substitution* $[\nu(a_1)/a_1, \ldots, \nu(a_k)/a_k]$*, where* $X = \{a_1, \ldots, a_k\}$*. Let* $(-)^{\mathrm{nov}(X)}$ *stand for* $(-)[\nu]_X$ *where* $\nu : X \to \mathbf{A}$ *is novel.*

Definition 21 (Substitution)**.** *Let* $Z(T, a) = \{u \in U_T : u \lhd_T a\} \cup \{e \in E_T : e \lesssim_T a\} - \{a\} = \{a_1, \ldots, a_k\}$*. For* $\mathbf{A}(\Gamma) \cap \mathbf{A}(\Theta) = \emptyset$*,*

$$\mathrm{Sub}^T_{\langle t,a\rangle}(\Gamma^\Theta_I) = \langle \Gamma[\nu_1]_{Z(T,a)}[t/a], \Theta, [T \rtimes ([T]_{Z(T,a)}[\nu_1]_{Z(T,a)})], I[\nu_1]_{Z(T,a)}[t/a]\rangle,$$

where $\nu_1 : Z(T, a) \to \mathbf{A}$ *is novel.*

Definition 22 (Merge).
For either $\mathbf{A}(R) \cap \mathbf{A}(S) = \emptyset$ *or* $[R]_{\Delta\cup\Psi} = S$*,*
$\Gamma^\Theta_{RI}[\Delta^\Psi_{SJ}]^M = \bigoplus^{R\bowtie S}_{\gamma\in\Gamma} \Big(\{\gamma\}^\Theta_{RI} \otimes^{R\bowtie S} \Delta^\Psi_{SJ} \otimes^{R\bowtie S} \bigotimes^{R\bowtie S}_{\langle t,u\rangle\in M'_{IJ}(\gamma)} \mathrm{Sub}^{R\bowtie S}_{\langle t,u\rangle}(\Delta^{\{0\}}_J)\Big)$,
where $M'_{IJ}(\gamma) = \{\langle t, u\rangle \in M_{IJ} : u \in U_S \wedge \exists\psi_{\in\Psi}(\psi[t/u] \subseteq \gamma \wedge \psi \not\subseteq \gamma)\}$